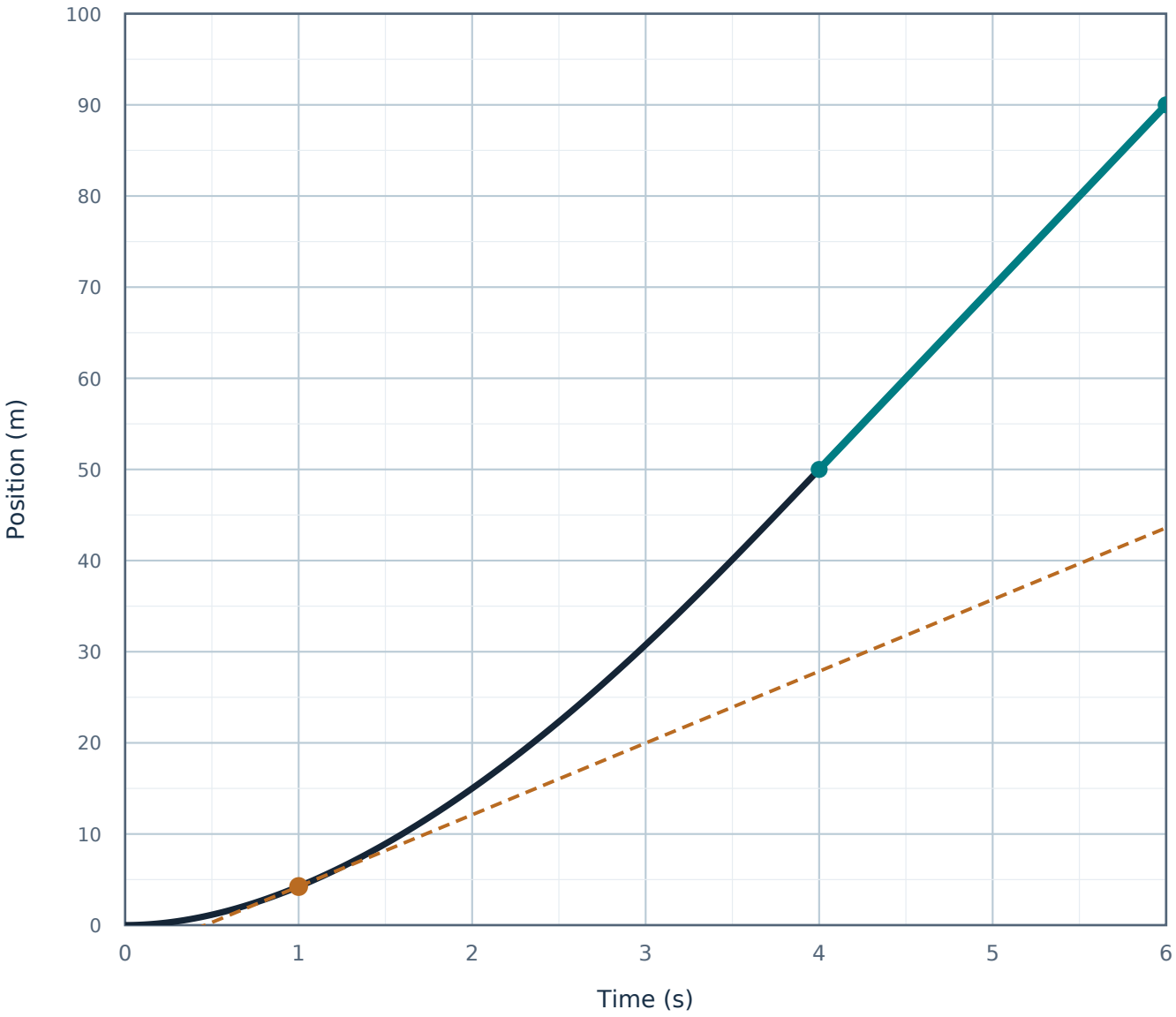
Position-Time Graph: Extension Activity

Grade 10 / MYP 5 | Worked solutions

Orange dashed: tangent at 1 s. Teal: constant-velocity section, 4-6 s.

The tangent is drawn from the exact constructed curve.



A. Velocity at t = 1 s

Gradient of tangent at 1 s ≈ 7.9 m/s (constructed value 7.87 m/s).

B. Maximum velocity

$v = \Delta x/\Delta t = (90 - 50)/(6 - 4) = 20$ m/s.

C. Description of the motion

Initially stationary; position increases at an increasing rate.

Velocity increases, while acceleration gradually decreases to zero.

From 4 to 6 s, velocity is constant at 20 m/s (terminal speed).

Teacher note: an idealised smooth model, not an exact aerodynamic drag law.

Initial acceleration is 10 m/s²; terminal behaviour begins at 4 s.

In a real fall, the terminal velocity is approached progressively.

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